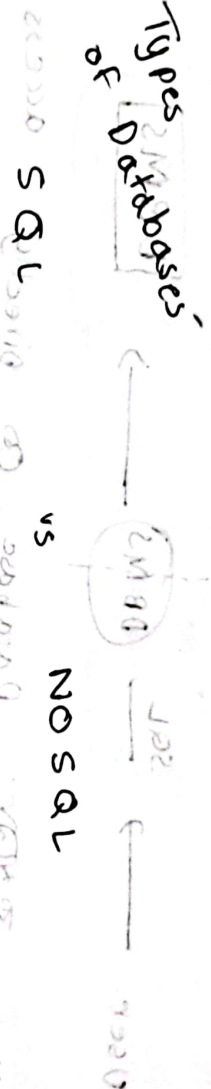


DBMS → Database 93 3478 Covering (EMAG)

Types of Databases



Table

Relational Database

Document, Key-Val, etc.

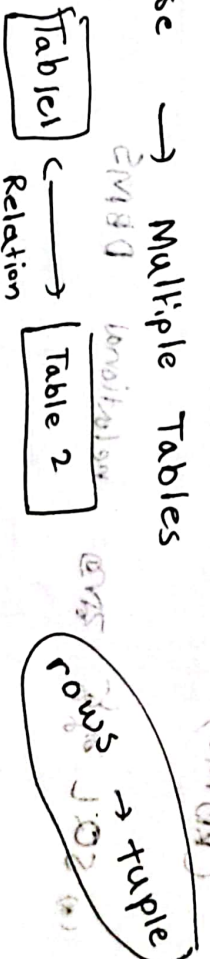
MySQL

SQL → Structured Query language

Programming language used to interact with relational databases.

MySQL server

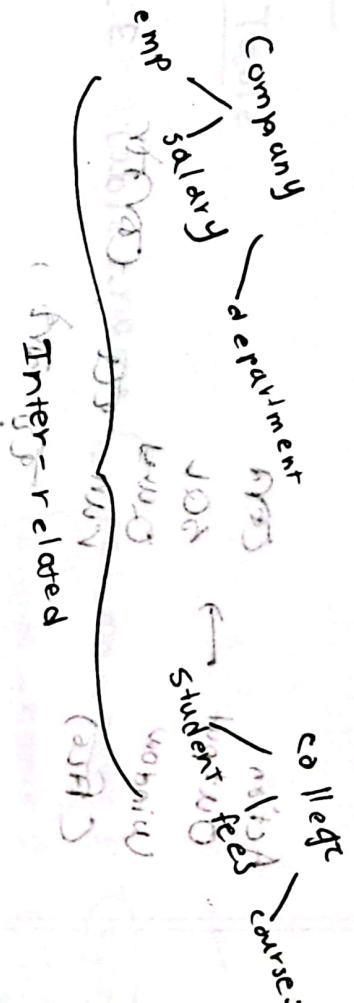
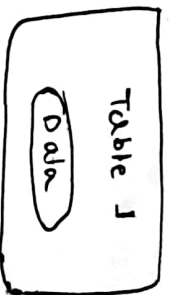
Database → Multiple Tables



col → design (Schema)

row → Single entity 93 3478 information.

Database



columns → structure / schema

for store data

Creating our First Database

First SQL Query: `CREATE DATABASE db_name`

IF A database is not exist

`CREATE DATABASE db_name`

`DROP DATABASE db_name`

→ `CREATE DATABASE IF NOT EXISTS`

(2)

Admin Panel

Admin Tab

Schem a Tab

1 user

Action
Output
Window
(File)

Can → SQL Query run - center
- center

End y ;



Database run

Database run



Highlighted option run

Sys Already query Default. then no problem

Database Created then 3030000 schema

right-click → Refresh All

→ then Database then 10 Table 10000

Delete

any then DROP DATABASE

DELETE FROM DATABASE

USE Temp1;

Active Database

SQL Queries

Creating First Table

CREATE TABLE table_name (column_name 1 type constraint
column_name 2 type constraint
column_name 3 type constraint

); [2 + 1 + 1] 1 pt. marks

USE db_name

id	name	age

constraint

↓
Data
more
after
3rd fe
constraints.

MySQL Workbench

Setting up connection (+)

→ & Password for the server

SQL Additions

Sys ~~SQL~~ By default database

Tab Hidding & 3 option for.

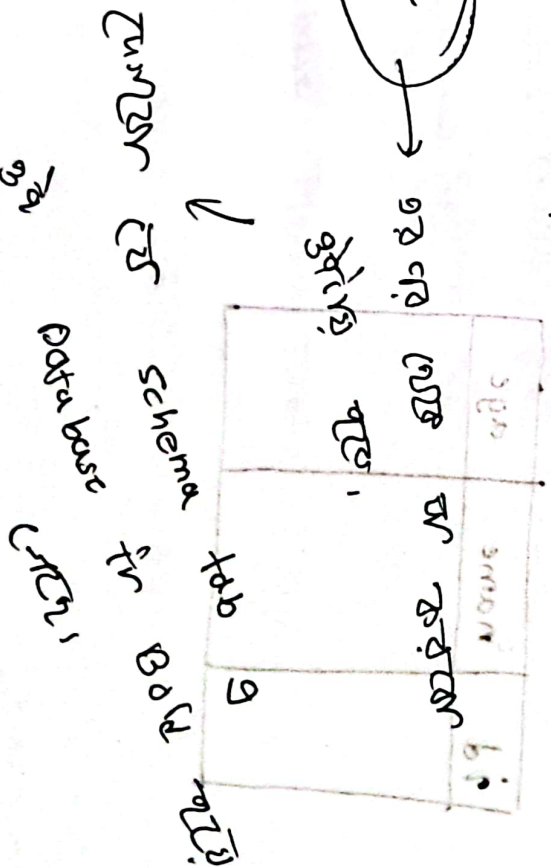
→ Classroom.sql [Ctrl + S]

Schema 50 યાદ Database દાખલ થાય છે

right click + Refresh All

જુ Just highlighted part for run કરાવવા.

Use Database



VARCHAR (50)

varchar 100

Data Type

Constraint

Table PRIMARY KEY & foreign key

Table & Table & Option & under & character

Table Created.

Inserting Data into Tables

INSERT INTO Table (col1, col2, col3) VALUES (val1, val2, val3)

Primary key & Unique → Table NO. primary

Table

Viewing

Table → cols → icon
SELECT * FROM table

(5) T18

SQL Datatypes

(002) 2A113 2A1V

CHAR

VARCHAR

string (0-255)

1987 2100

256 size of string

fixed length

string to store

0-255

1987

or for 255 bit or 255 present

BLOB (0-65535)

INT

TINYINT (-128 to 127)

BIT → can store x-bit values

X can range from 1 to 64

BIT (2)

2-bit of

number values
store
fixed bit

Bit (1)

↓

0, 1

{ 00, 10
01, 11 }

FLOAT → can't
DOUBLE → can't

Bit (2)

TINY INT - BOOLEAN

Signed & Unsigned

Range 0 to 255 use where

TINY INT UNSIGNED (0 to 255)
TINY INT (-128 to 127)

Type of SQL Commands

DDL

Definition

Creation, deletion

DDL

Manipulation

DQL

Query →

Query →

select

DDL
CREATE, ALTER, DROP, TRUNCATE

DML
INSERT, UPDATE, DELETE

pg
grant & revoke
permission:
{ '1', '00' }

MAJOOB - TWT PUTT

Database related Queries

અનફર ડટા Database બનતો બુટ સુરે
અનફર ડટા બન્યો બન્યો Created અનફર
અનફર

IF NOT EXISTS

Turning
અનફર બન્યો 102 10 3901

અનફર Use અનફર Error અનફર

DROP → IF EXISTS

SHOW DATABASES
SHOW TABLES

ડટા Database use
અનફર ડટા અનફર
અનફર ડટા

Table Related Queries

Primary key

— not null
— unique

SELECT * FROM table name

INSERT INTO table name (col1, col2) VALUES

(col1-v1, col2-v2)

(col1-v2, col2-v2);

-v1
v1

1st Row

Basically

v2

2nd Row

Basically

(row1, row2), (row3, row4), (row5, row6)

So, Insert INTO table name (col1, col2)

VALUES (row1), (row2), (row3), (row4), (row5)

(row5);

Practice Qs 1

company named → XYZ

DB → Table { ^{id}name } Create Table Y (

salary id)

create ~~xyz~~ database xyz

use xyz;

Table (name, salary, ...)

Table

Data Type Constraint

name type constraint

primary

foreign

unique

Database

Table Creation

Load Data

Insert Data

(col1, col2, col3) values (row1, row2, row3)

Select * from xyz

where (col1 = 'value1')

order by

Key

Primary key (Unique) → Null not allowed

Foreign key

Set of columns that refers to the primary key of another table.

Each table can exist without a primary key.

A table can have multiple primary keys & null values.

Constraints.

NOT
NULL

UNIQUE

PRIMARY
KEY

↓
all values
in cols
are different

Primary key ~~अवश्य~~ 3 Use करे थे

col x, col y, col z, primary key [x]

Foreign key $\rightarrow$ ~~दो~~ Table के Link करे

Foreign key (~~ए~~ col)

~~अवश्य~~ foreign key, ~~अवश्य~~

references

* S Table

अवश्य Col 92

(S Table ~~अवश्य~~ Primary key)

x references

YAN

ADDITION

TOM

JUN

DEFAULT $\rightarrow$ set default value of x col.

$\hookrightarrow$ by default after value set ~~अवश्य~~

col type default x

अवश्य FROM

col type constraint

Check $\rightarrow$ it can limit the values allowed in a column
 $\hookrightarrow$ value add to the table check to the table

col type

constraint col type check (a condition)

DISTINCT

$\rightarrow$ Only Unique values

Where?

Define some conditions

$\hookrightarrow$ Extra condition, col type
 $\rightarrow$ extra condition 3 Full-fill 2 rows

Select col 1, col 2 from table-name
Where conditions;

operator

where

Condition 1

and / or

Condition 2

Operators

where Clause $\rightarrow \checkmark$

+, -, *, /, %

=, <, >, <=, >=, <>, <=, >=

AND, OR, NOT, IN, BETWEEN, ALL, LIKE, ANY

2, 1

Between

Center range

or and for select

area

Between 80

AND 90

Inclusive

IN
matrices

any

value

in list

where

or in (a, b, c)

NOT IN ✓

Condition for false return Negative to false.

Limit

row limit return false.

Select * from student

limit 3;

3rd student
of data of
return.

Order By Clause

Sorting on basis

Ascending
Descending

Order by

col-name

ASC;

/DESC

AGGREGATE Functions

perform a calculation on a
set of values and return a
single value.

COUNT()
MAX()
MIN()
SUM()
AVG()

SELECT FUNC (col)
FROM table-name

Group By

→ Group Rows

as center col of Basis of Group answer

→ Multiple summary groups

Group By or other Aggregate function
use as

Select city

→ col of basis of type

From Student

Group by city

→ Group formation
← Form

Select

(x, y, z)

} at the end of column

Group by (a, b, c)

(col of basis)

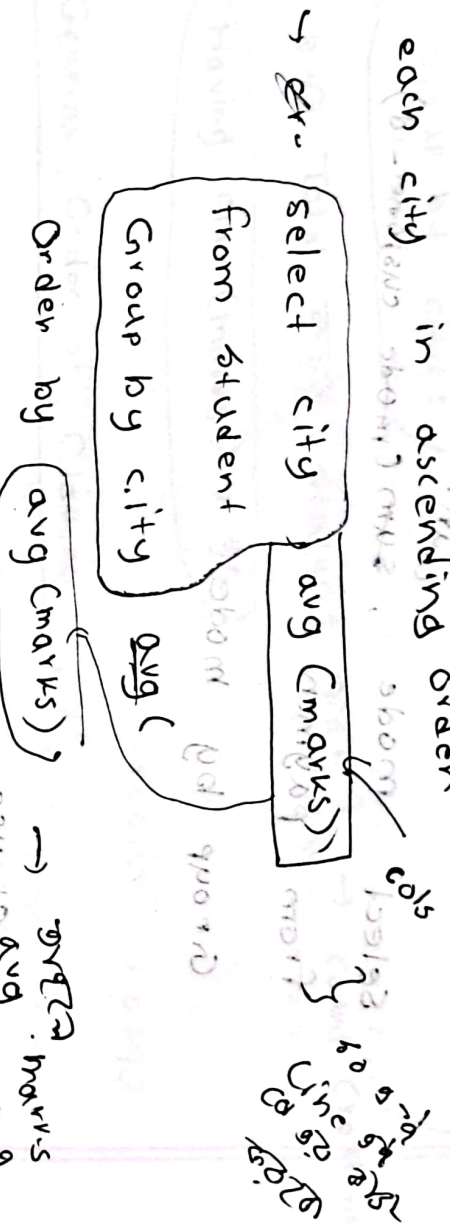
(*) Group By or first thought was Group

for faster

basis of answer
(can col)

Practice Q's

1) Write the query to find avg marks in each city in ascending order



Composite
primary key $\Rightarrow$ avg(marks)

single row wise $\Rightarrow$ city

1) Total payment method

customer id	customer	mode	city
1	200001	Credit	Net.
2	200002	Credit	Net.
3	200003	Credit	Net.
4	200004	Credit	Net.
5	200005	Credit	Net.
6	200006	Credit	Net.
7	200007	Credit	Net.
8	200008	Credit	Net.
9	200009	Credit	Net.
10	200010	Credit	Net.

total payment according to each
Payment method? (mode)

Select mode, sum(mode, customer-id)
from payment

Group by mode

Having clause

Similar to where
applies condition on some rows.

(x) Grouping of rows condition apply
ଅଟେ ଯେଉଁଠି → Having

(x) Count no. of students in
each city where max, marks cross 90

Where → rows ଯି ଡାଟା ଥାଏ

Having → Groups ଯି ଡାଟା ଥାଏ

select city
from student
group by city;

→ Group rows

or Total rows, student name → can't roll no

Having max (marks) > 90;

General Order of Clauses

Select

col

From table-name

WHERE CONDITION

GROUP BY col

HAVING condition

Order by cols ASC;

where

rows

having

groups

where

Group By

Having

10 = 2376090 - 3112 - 102 = 132

Table Related Queries

Update → updates existing rows/table

Set col 1 = val 1, col 2 = val 2
where condition;

Update student

Set grade = '0'

where grade = 'A';

Student Table

grade = '0'
set grade

grade = 'A' idol



change data

Safe Mode Off करें

SET SQL - SAFE - UP DATES = 0;

Delete

(to delete existing rows)

Delete from table-name
where condition;

Revisiting Foreign Key

Primary → bridge

courses

id	name
101	science
102	English
103	Space

Teacher (FK)

101
102
103



(*) primary key is FK always

(*) Foreign Key (dept id) references

Foreign key (col) REFERENCES
primary (primary col)
key of table

always last of table,

MySQL Workbench



Database



Reverse Engineer



Continue



F+R
Diagram

Use Table as Master Foreign Key for

use table as Child Key/ Table

Master Parent Table

Cascading for FK

Cascading → as as table for change etc,

as as table as change -> as as table

as as table as change



as as table as change

as as table as change

as as table as change

Truncate

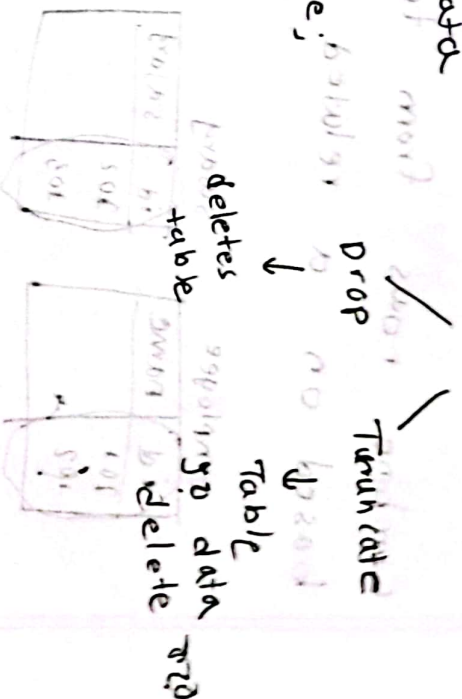
✓ deletes table's data

Truncate table table-name;

Update student

Set grade = "O"

WHERE grade = "A"



Practice Qs

(a) Update Col to new Rename table

ALTER → CHANGE (i) TABLE

CHANGE old New Type, (ii) name

(b) Delete all the students who scored marks less than 80.

(c) Delete the col for Grades

ALTER TABLE

Joins in SQL

Combine rows from two or more tables, based on a related col between them

employee		salary	
id	name	id	salary
101		102	
102		103	

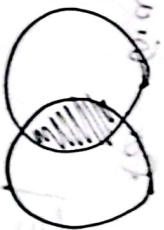
Types of Joins

- (i) Inner Join
 - left
 - right
 - full
- (ii) Outer Joins

Inner Join

→ get table of common after carrier
 select (Inner Join) not rest return

Left Join



101 not salary 102

Joins in SQL

Combine rows from two or more tables, based on a related col between them

employee		salary	
id	name	id	salary
101		102	
102		103	

"O" = outer join
"A" = inner join

Types of Joins

- (i) Inner Join
- (ii) Outer Joins
 - Left
 - Right
 - Full

Inner Join

→ Both table's common column after connect
syntax (Inner Join) and result

Left Join



LEFT JOIN

Left Join



Right Join



Full Join



L

id	name
101	✓
102	✓

R

id	salary
101	
102	
103	

} Left Join result

Right Join → 101, 102, 103 (Right side of table)

Full Join → Both Table of

Inner Join

SELECT col

FROM table A

INNER JOIN table B

ON table A.col-name

= table B.col-name;

[Table A is first then col after
joins then compare after table
B is given]

ALIAS
A V's Name is the same as the
way

Fall Data, 7

102	101	100
-----	-----	-----

105	✓	100006
100	✓	
100		

Retains all records when there is a

on right table.

on right table.

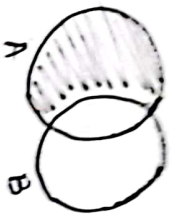
June 2010

Unit Full
Jorn 132

On page 101-102 of the book "The Book of the Dead" by E. A. Wallis Budge, the following is written:

June 18 88

Left Exclusive Join



Left Join

on $a.id = b.id$

Where $b.id$ is Null



જિન બાજુ



$b.id$ not null & બાજુ (Left Join $\rightarrow y.null + y.not$)
of x

+
→ $b.null$ હોય તો

બાજુ, નો, બાજુ

$b.null$ નો બાજુ.

અને Opposite નો $o.null$ બાજુ.

Self Join

Regular Join ની જેમ જોડવા માટે connect
કોણે જોડવાનું હોય તો જોડવા માટે
Join કરવા પડે.

Right Exclusive Join



Full Exclusive Join



Syntax of Self Join

SELECT columns (

FROM table as a

JOIN table as b

ON a.col-name = b.col-name

Same table join $\rightarrow$ self join

One table $\rightarrow$ self join

multiple $\rightarrow$ self-join

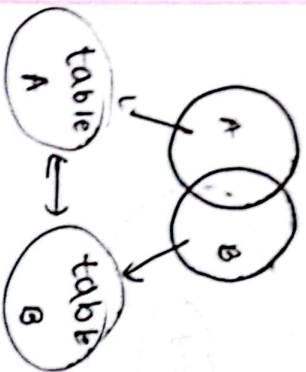
Union

Combine the result-set of two or more

select

statements

SELECT



दोनों स्टेटमेंट्स And

Then दोनों के unique value after filter.

दोनों select statements combine

(i) Every select

should have same no. of Col.

(ii) cols must have similar data type

(iii) cols in select should be in same order

SELECT column(s) FROM table A

UNION

SELECT Column(s) FROM table B

UNION -> full

join

UNION ALL

Duplicates also

use

Sub Queries / Nested Query

Queries are queries with other queries
Queries are queries like 1, 2, 3, 4

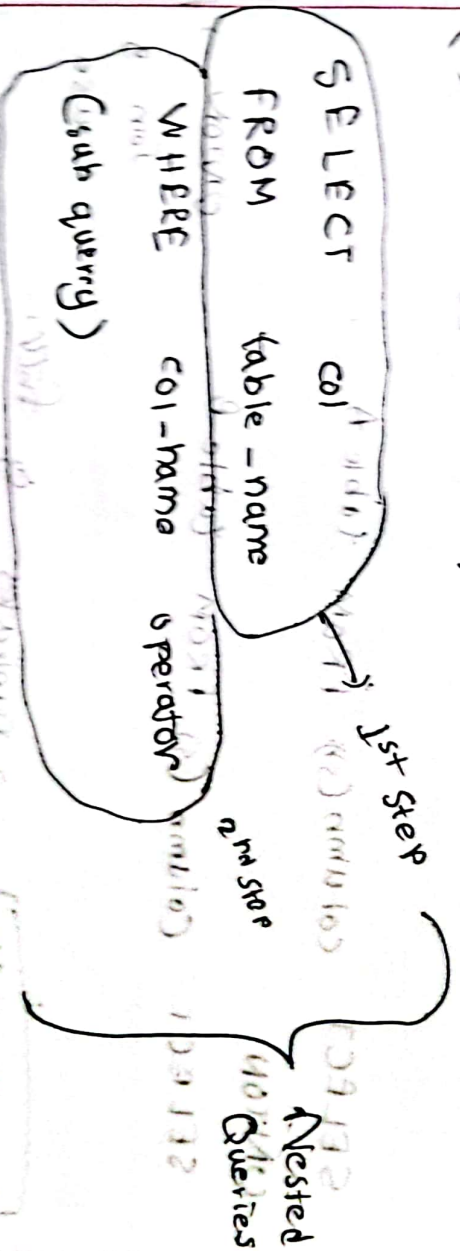
Select statements

[[]]



3 type of query :-

- (i) Select query
- (ii) From query
- (iii) Where query



જાતી Table છે જાન ત્રી step થી calculation
કરે પણ થાય → SQL Sub Queries

Average
કરે કરે કરે → AGGREGATE

1st step થી કરે કરે answer થી કરે કરે
use કરે 2nd step 1. sub Queries થી કરે કરે,

Dynamic → જાતી change કરે કરે કરે કરે
કરે કરે effect કરે કરે.

so, subquery.

→ Data change
કરે કરે change
કરે કરે કરે,

SELECT name, marks
FROM STUD, student

1st step

WHERE marks >

(SELECT avg (marks) FROM student);

1st Step

WHERE Operator (1st step)